

## Standard Operating Procedure

|                                                                                                                          |  |                                    |  |                              |  |                                  |  |                             |  |            |  |
|--------------------------------------------------------------------------------------------------------------------------|--|------------------------------------|--|------------------------------|--|----------------------------------|--|-----------------------------|--|------------|--|
| s. Milestone Gears Pvt. Ltd.,<br>Unit - 4, Opp. Himachal Fibers Ltd.,<br>d. Area, Barotiwala, Distt. -<br>Jalandhar (HP) |  | OPN. NO.<br><b>220</b>             |  | पिछला आपरेशन<br>अगला आपरेशन  |  | Shot Blasting<br>Gear Chamfering |  | MACHINE NO.<br>REVISION NO. |  | Lab.<br>00 |  |
| Assy. Gear 3rd C-Mesh<br>1512A01401                                                                                      |  | OPERATION<br>Metallurgical Testing |  | APPD. BY: <i>[Signature]</i> |  | REVISION DATE                    |  | DATE OF ENACTMENT           |  | 25/11/13   |  |
| CUST. NAME<br>MATERIAL                                                                                                   |  | TAFE(8x8)<br>16MnCr5               |  | CHD. BY: <i>[Signature]</i>  |  | FIXTURE NUMBER :                 |  | TOOLS / CUTTERS DETAIL :    |  |            |  |

## SPECTION INSTRUMENTS DETAIL :

Rockwell Tester, Vicker H. Tester, Microscope

| NO. | MACHINE / EQUIPMENT PARAMETERS    | VALUE | LOCATE | CLAMP | REST | INSPECT | MAJOR Mj | MINOR Mn | CRITICAL Cr |
|-----|-----------------------------------|-------|--------|-------|------|---------|----------|----------|-------------|
| 1.  |                                   |       |        |       |      |         |          |          |             |
| 2.  |                                   |       |        |       |      |         |          |          |             |
| 3.  |                                   |       |        |       |      |         |          |          |             |
| NO. | JIG / FIX. / CTG. TOOL PARAMETERS | VALUE |        |       |      |         |          |          |             |
| 1.  |                                   |       |        |       |      |         |          |          |             |
| 2.  |                                   |       |        |       |      |         |          |          |             |
| 3.  |                                   |       |        |       |      |         |          |          |             |

## CONTROLLING OF DIEMENSION

| No  | Dimension                         | Specifications | Instrument       | Operator         | Freq.            |
|-----|-----------------------------------|----------------|------------------|------------------|------------------|
| 1A. | Previous Operation                | Shot Blasting  | Visually         | 100%             | 100%             |
| 1.  | Surface Hardness                  | 58~63HRC       | Rockwell Tester  | 1 piece per heat | 1 piece per heat |
| 2.  | ECD (Cut off HRC)                 | 0.76~1.14mm    | Vicker H. Tester | --do--           | --do--           |
| 3.  | Core Hardness                     | 27~41HRC       | Rockwell Tester  | --do--           | --do--           |
| 4.  | Surface Carbon Content            | 0.80~1.10 %    | Microscope       | --do--           | --do--           |
| 5.  | Surface Hardness on Internal Bore | 27HRC Min.     | Rockwell Tester  | --do--           | --do--           |

Note:-

Rejection होने पर Machine को तुलन रोक दें और shift incharge और Q.C. Insp. की मदद से SPOT ANALYSIS करावें उसके बाद ही Machine को चलाएँ।

ENGG. CHANGES.

## OPTN. SKETCH



CORE HARDNESS  
MEASURED AT ROOT DIA.  
6.4 FROM END OF TOOTH

ECD & HARDNESS  
MEASURED AT MIDPOINT  
OF TOOTH PROFILE 6.4  
FROM END OF TOOTH.

CONTROLLED COPY

SR. NO.

01

ISSUED TO

NOTE:-

मशीन चलाने से पहले ऑपरेटर पिछला आपरेशन चेक करेगा।  
पिछला आपरेशन होने के बाद ही अगला आपरेशन करेगा।

25/11/24

## WORKING INSTRUCTIONS

Metallurgical Testing के लिये नीचे लिखे Instruction को Follow करें :-

- 1) Rockwell Hardness Tester में Component की Surface Hardness चैक करें।
- 2) Component को काट कर एक सैम्पल चैक करें।
- 3) Rockwell/Vickers Hardness Tester में Sample की Core Hardness चैक करें।
- 4) Vickers Hardness Tester में Sample की ECD चैक करें।
- 5) Nital से Sample को Etching करें।
- 6) Etching किये हुये Sample का Case & Core में Microstructure Microscope द्वारा चैक करें।